Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

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1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

`https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist`

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.
The labels are not required as reconstruction targets. The original image is the target:

$$\boxed{x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

```
Original image
Reconstructed image
Reconstruction error
```

For the denoising model, display:

```
Original clean image
Noisy image
Denoised reconstruction
```

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.
The encoder maps the input to a lower-dimensional representation:

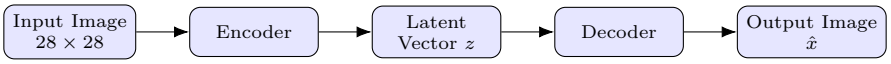
$$z = f_\theta(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_\phi(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2.$$



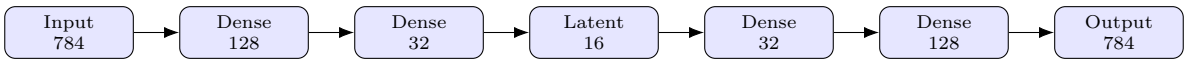
The latent vector is a compressed representation of the input.

7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.
Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.
Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

The fully connected autoencoder was trained for 20 epochs with a 16-dimensional latent representation. The reconstruction figure shows that the main digit structure is retained, while thin strokes and fine boundary details are less accurately reproduced.

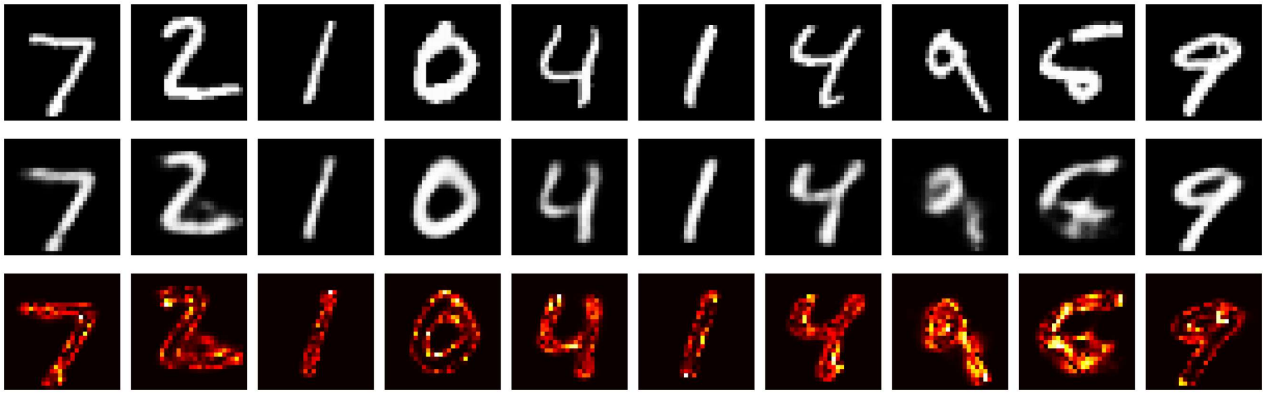


Figure 1: Original images, fully connected autoencoder reconstructions and absolute reconstruction errors.

The plot shows that the autoencoder captures the dominant shape of each digit, but some reconstructed strokes are smoother or weaker than the originals. The reconstruction error is concentrated around digit boundaries and fine strokes because the latent representation compresses the 784-dimensional input into only 16 values. Overall, the model learns the major image structure, but the reconstruction is not able to preserve every local detail.

Plot 2: Training and Validation Reconstruction Loss

The final training loss after 20 epochs was approximately 0.1050, while the validation loss was approximately 0.1061.

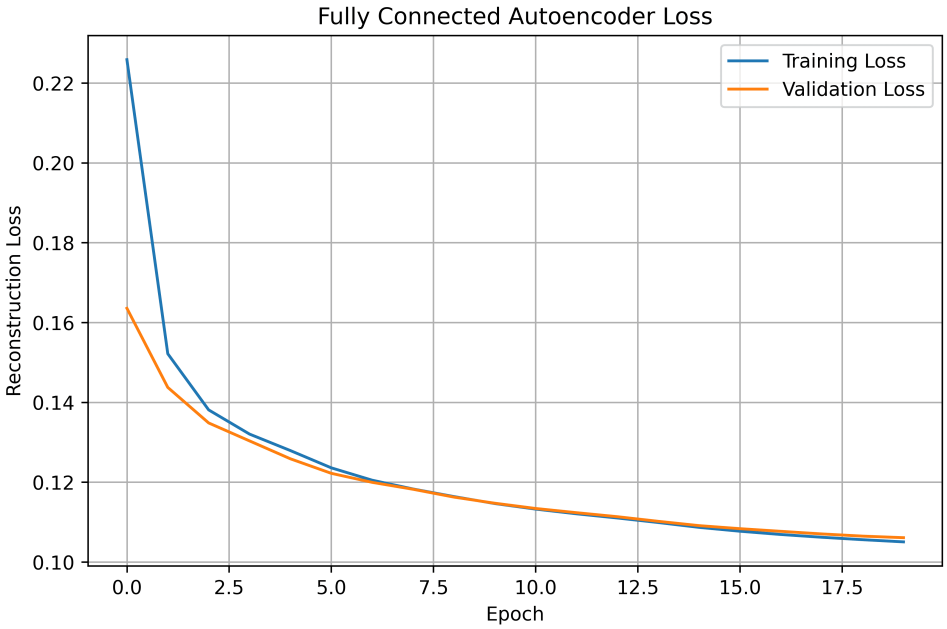


Figure 2: Training and validation reconstruction loss of the fully connected autoencoder.

Both curves decrease steadily throughout training and remain close to one another. This indicates that the model converges and that there is no strong evidence of overfitting in the recorded training history. The small gap between training and validation loss suggests that the learned mapping generalizes reasonably well to unseen validation images.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2.$$

For the fully connected autoencoder:

$$\boxed{MSE = 0.014010}$$

Mean Absolute Error

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|x_i - \hat{x}_i|.$$

For the fully connected autoencoder:

$$\boxed{MAE = 0.042117}$$

Structural Similarity

The mean SSIM over the test set was:

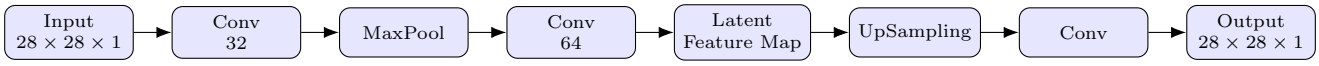
$$\boxed{SSIM = 0.843624}$$

Metric	Value
Test MSE	0.014010
Test MAE	0.042117
Mean SSIM	0.843624

The MSE and MAE quantify pixel-level reconstruction error, whereas SSIM is more sensitive to structural similarity. The results indicate that the basic fully connected model reconstructs the general digit structure but loses some fine spatial detail.

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.
The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

The convolutional autoencoder preserves the two-dimensional image structure, whereas the fully connected model first flattens the image into a 784-element vector.

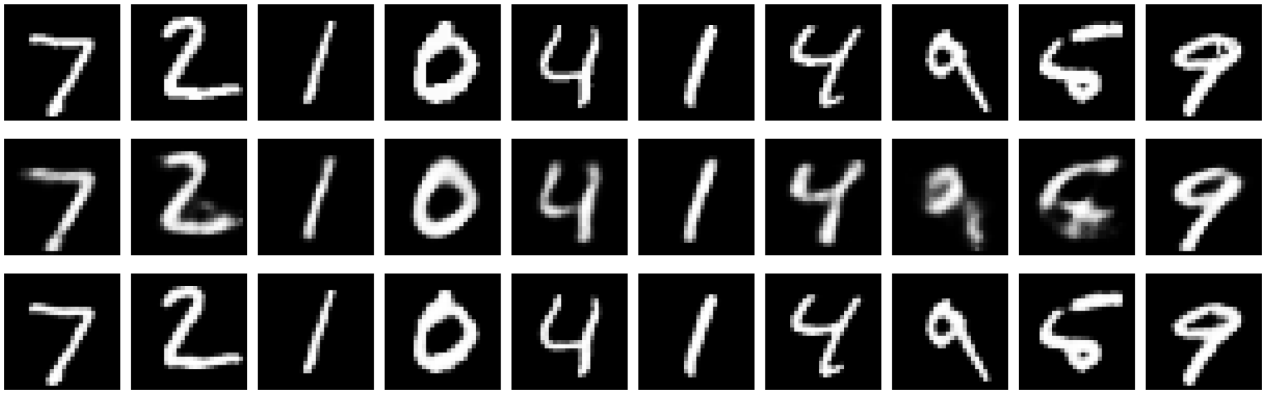


Figure 3: Comparison of fully connected and convolutional autoencoder reconstructions.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.014010	0.042117	0.843624	211040
Convolutional AE	0.001133	0.010079	0.988807	74497

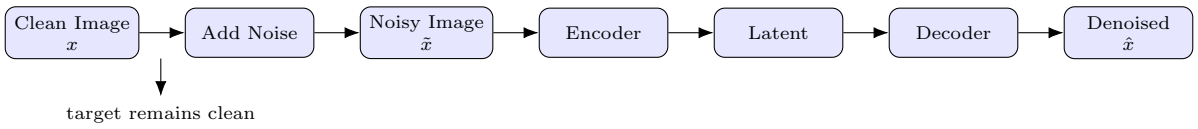
The convolutional autoencoder produces substantially lower MSE and MAE and a much higher SSIM. It also uses fewer parameters than the fully connected model in this experiment. The main reason is that convolution operates locally and reuses filters across the image, allowing the network to exploit spatial locality and translation-related patterns. The fully connected model treats all pixels as independent input positions after flattening, so it has less direct access to local spatial structure.

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.
The training mapping is:

$$\boxed{\tilde{x} \to \text{Encoder} \to z \to \text{Decoder} \to \hat{x}}$$

where $\tilde{x}$ is a noisy version of the clean image x.



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \qquad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.
Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

The denoising autoencoder was trained using Gaussian noise with $\sigma = 0.2$, while the clean image remained the reconstruction target.

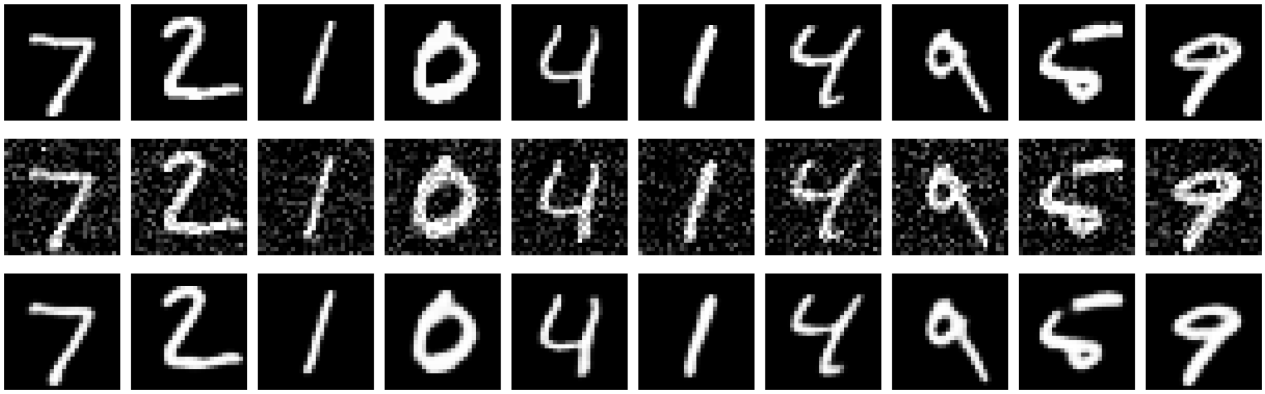


Figure 4: Clean, Gaussian-noisy and denoised test images.

For $\sigma = 0.2$, the denoising autoencoder obtained:

$$MSE = 0.003012, \qquad MAE = 0.016943, \qquad SSIM = 0.962498.$$

The noisy images contain random intensity perturbations, while the denoised reconstructions recover the overall digit strokes and remove a large portion of the corruption. The high SSIM shows that the important spatial structure is preserved. Some small variations remain because the model must infer the clean image from a corrupted observation rather than simply copying the input.

Plot 5: Noise Level versus Reconstruction Quality

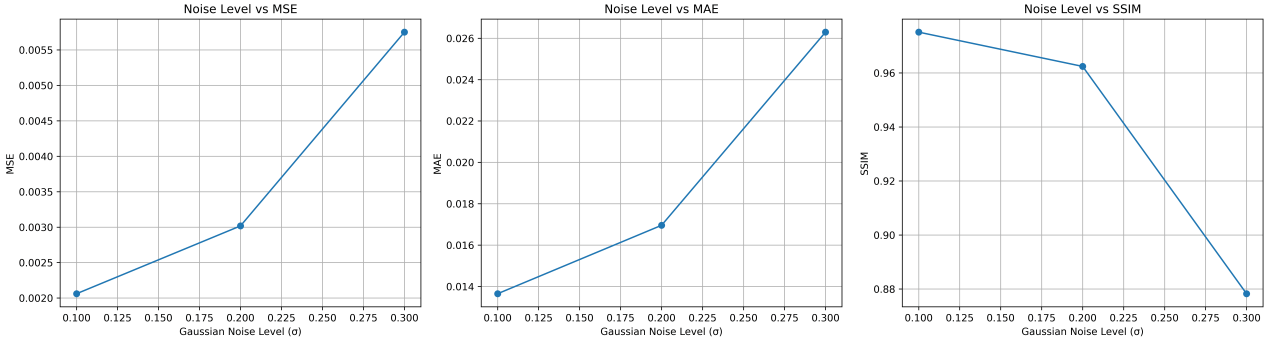


Figure 5: Effect of Gaussian noise level on MSE, MAE and SSIM.

Noise Level σ	MSE	MAE	SSIM
0.1	0.002061	0.013650	0.975110
0.2	0.003017	0.016957	0.962421
0.3	0.005750	0.026295	0.878240

Increasing Gaussian noise causes MSE and MAE to increase, while SSIM decreases. At $\sigma = 0.1$, the model retains high structural similarity, but at $\sigma = 0.3$ the stronger corruption makes recovery more difficult. The trend occurs because higher noise introduces larger deviations from the clean pixel values and removes more information that the decoder must reconstruct. Thus, denoising quality degrades as the corruption becomes more severe.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

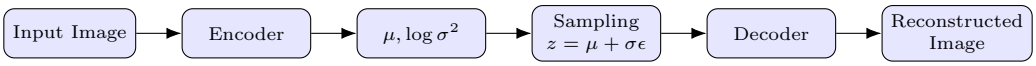
$$q_\phi(z|x) = \mathcal{N}\left(\mu(x), \mathrm{diag}(\sigma^2(x))\right).$$

The encoder produces:

$$\mu, \qquad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}\left(q_\phi(z|x) \parallel p(z)\right) = -\frac{1}{2}\sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.
Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

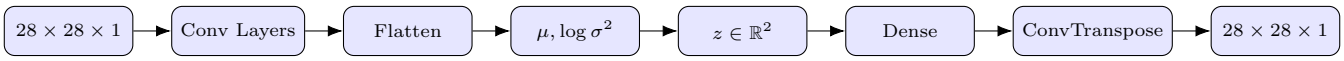
The encoder should produce:

$$\mu \in \mathbb{R}^2, \qquad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \to 28 \times 28 \times 1.$$

A compact architecture is:



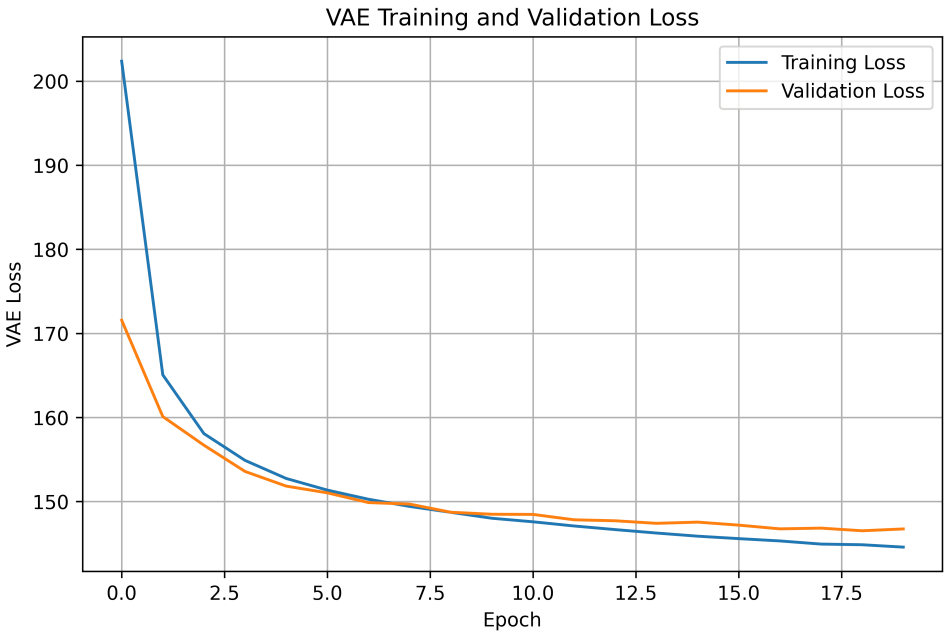


Figure 6: VAE training and validation loss over 20 epochs.

The VAE loss decreases substantially during training and then levels off. The final training loss was approximately 144.5823 and the final validation loss was approximately 146.7445. The relatively small final gap indicates that the training process reached a stable region without a large late-stage divergence between training and validation behavior.

18. Exploring the VAE Latent Space

The VAE uses a two-dimensional latent representation, allowing every test image to be represented by a point in a two-dimensional plane.

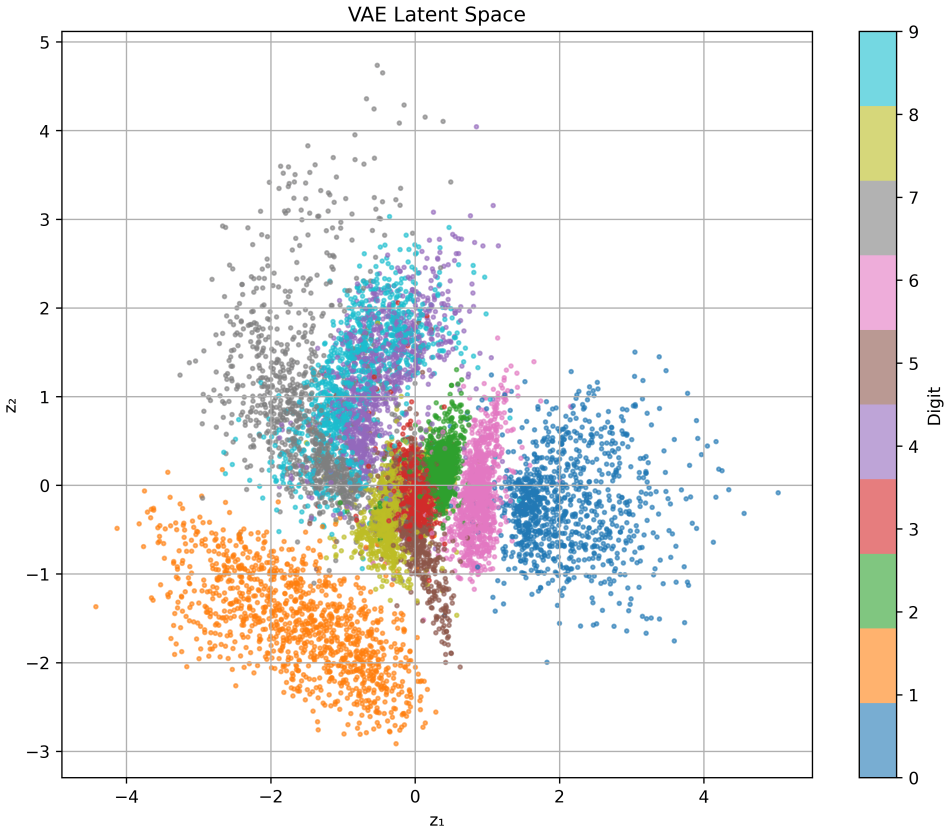


Figure 7: Two-dimensional VAE latent-space visualization, colored by digit label.

The latent space is not uniformly mixed: images belonging to similar digit classes tend to occupy related regions, although overlap is visible between several classes. This occurs because the encoder learns a continuous probabilistic representation that captures visual characteristics useful for reconstruction. The overlap is expected because handwritten digits can share similar shapes and the VAE is not explicitly trained as a classifier. The plot therefore shows both clustering structure and a continuous transition between different digit patterns.

19. Generating New Images with the VAE

Random latent vectors were sampled from a standard normal distribution and passed through the trained decoder.

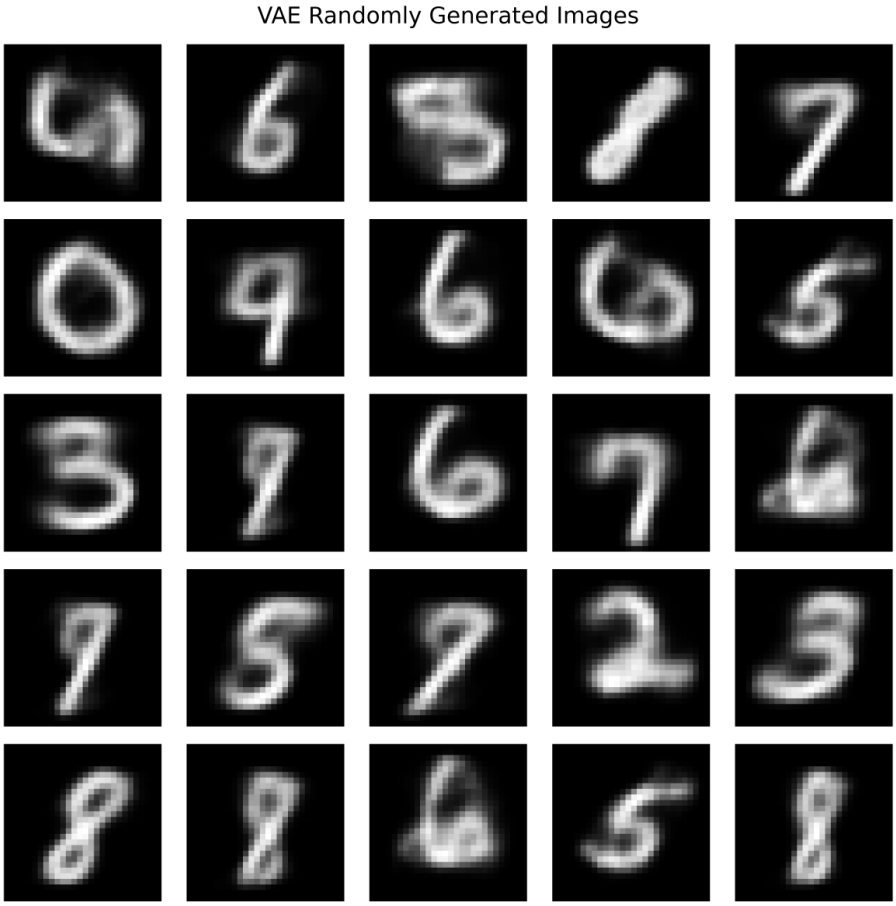


Figure 8: Twenty-five images generated by sampling random VAE latent vectors.

The generated samples resemble handwritten digits, showing that the decoder has learned a meaningful mapping from the latent distribution to the image space. At the same time, some samples are less sharp or ambiguous. This is expected because random samples can fall in latent regions that are between strongly represented training examples. The result demonstrates that the VAE is capable of generation rather than only reconstructing existing test images.

20. Latent Space Interpolation

Two latent vectors were selected and interpolated using

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \qquad \alpha \in [0, 1].$$

The experiment used 11 values of α from 0 to 1.

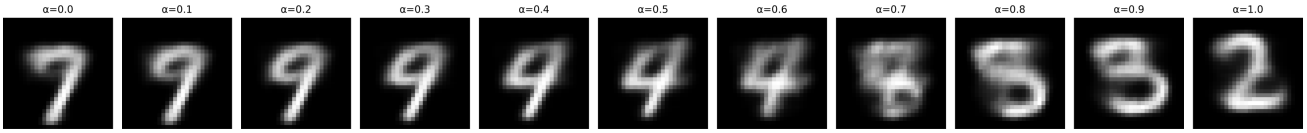


Figure 9: Latent-space interpolation between two VAE latent vectors.

The reconstructed images change progressively as α varies from 0 to 1. Rather than changing randomly between unrelated images, the decoder produces intermediate visual patterns. This smoothness is a useful property of the VAE because its regularized latent space is encouraged to be continuous and close to the chosen prior distribution.

21. Reconstruction Comparison Across Models

The final reconstruction comparison obtained from the executed notebook is:

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.014010	0.042117	0.843624	211040
Convolutional AE	0.001133	0.010079	0.988807	74497
Denoising CAE	0.003012	0.016943	0.962498	74497

The convolutional autoencoder gives the strongest deterministic reconstruction metrics among these three models. The denoising CAE has slightly higher error than the clean-input CAE because it must recover the clean image from corrupted input.

For the VAE, the losses and test metrics were:

VAE Metric	Value
Reconstruction Loss	140.211533
KL Loss	6.286471
Total Loss	146.498001
Test MSE	0.039331
Test MAE	0.093963
Mean SSIM	0.559854

The VAE objective is different from the deterministic autoencoders because its loss includes both reconstruction quality and a KL-divergence regularization term.

22. Reconstruction Error Distribution

For each test image, the reconstruction error was calculated as the mean squared pixel error:

$$e_i = \frac{1}{D}\sum_{j=1}^{D}(x_{ij} - \hat{x}_{ij})^2.$$

Across the 10,000 test images, the minimum error was 0.002156, the maximum was 0.161758, and the mean was 0.039331.

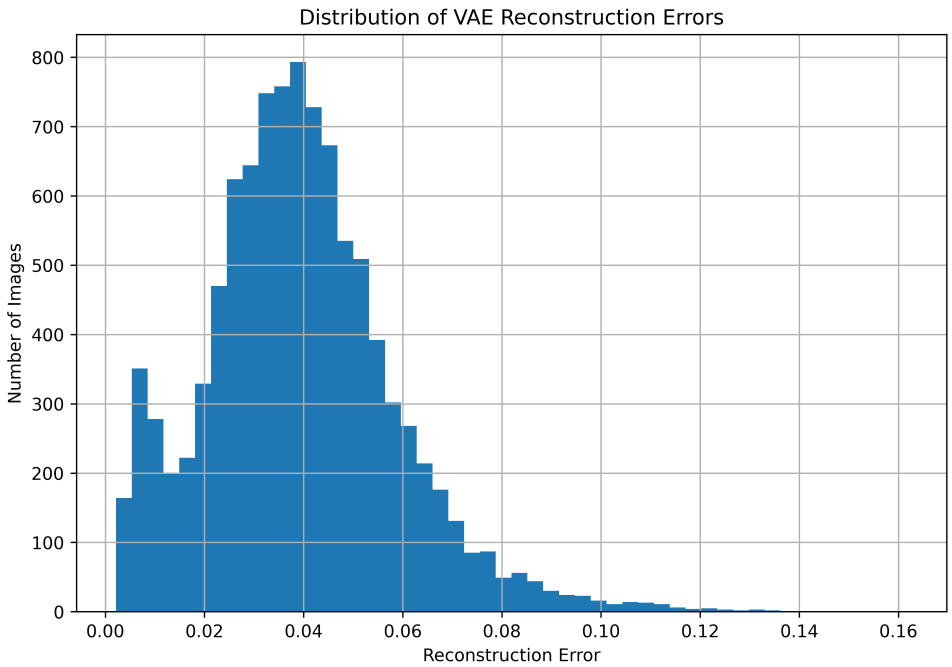


Figure 10: Distribution of per-image VAE reconstruction errors on the test set.

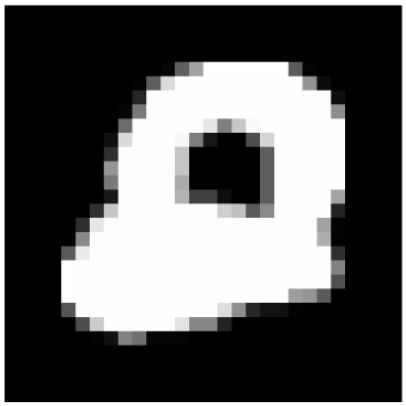
Most test samples have reconstruction errors in the lower part of the observed range, while a smaller number of images form a higher-error tail. The spread occurs because some handwritten digits are easier for the VAE to represent than others. Images with unusual stroke shapes, ambiguous writing or less typical structure can be farther from the dominant patterns learned by the decoder.

23. Required High-Error Image Analysis

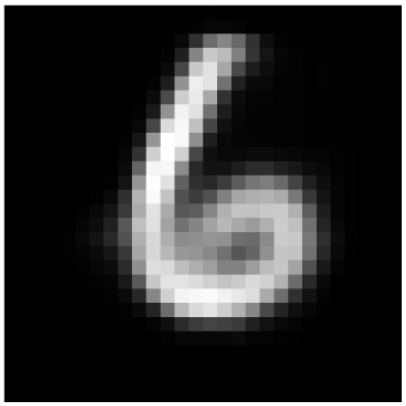
The five test images with the largest reconstruction errors were identified using the per-image VAE reconstruction error.

Sample	Test Index	Digit	Reconstruction Error
1	2462	2	0.161758
2	2326	0	0.145169
3	3727	8	0.140375
4	7871	8	0.135632
5	8	5	0.135311

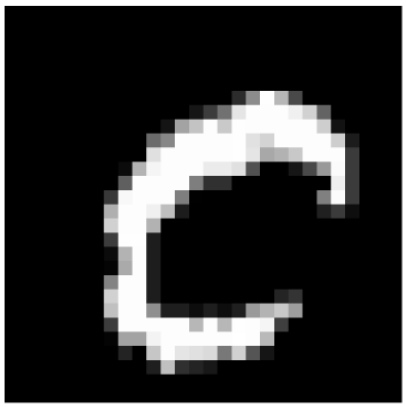
Original | Sample 2462



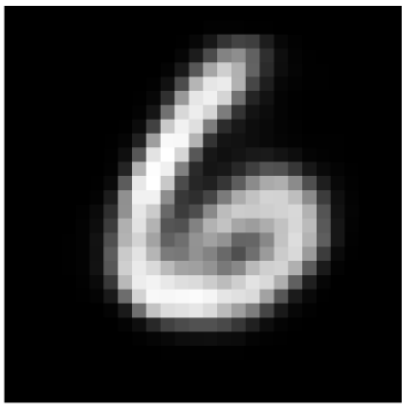
Reconstruction | Error: 0.16176



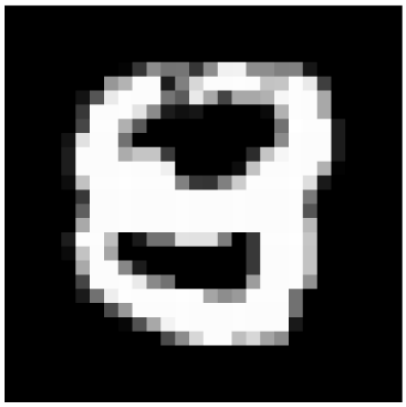
Original | Sample 2326



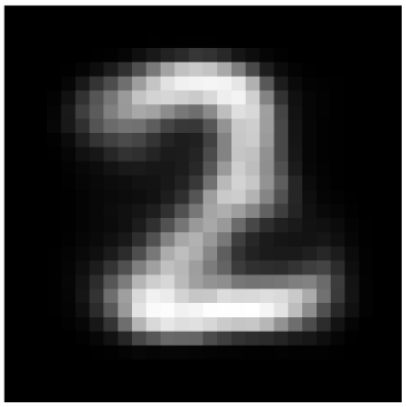
Reconstruction | Error: 0.14517



Original | Sample 3727



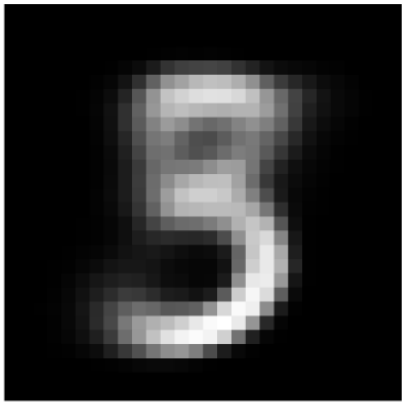
Reconstruction | Error: 0.14037



Original | Sample 7871



Reconstruction | Error: 0.13563



Original | Sample 8



Reconstruction | Error: 0.13531



These samples are difficult to reconstruct because their visual patterns are less well represented by the learned latent distribution. Digits with unusual proportions, ambiguous strokes or strong local variations can be mapped to a latent representation that does not preserve every detail. The high error therefore reflects a mismatch between the individual sample and the dominant image patterns learned during training.

24. Consolidated Results

The following values were obtained from the final timed evaluation in the notebook. The inference time is the measured prediction time on the test set for each model under that execution.

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.014010	0.042117	0.843624	211040	0.7576
Conv. Autoencoder	0.001133	0.010079	0.988807	74497	0.8233
Denoising CAE	0.003013	0.016950	0.962442	74497	1.3554
VAE	0.039306	0.093929	0.560105	133381	0.3529

The deterministic convolutional model achieves the lowest reconstruction error and highest SSIM in this evaluation. The VAE has a different objective because it must simultaneously learn useful reconstructions and a regularized latent distribution. Consequently, its reconstruction metrics should be interpreted along with its generative and latent-space properties rather than as a direct replacement for the deterministic reconstruction objective.

25. Required Inferences

Effect of latent dimension on reconstruction. Increasing the latent dimension generally improves reconstruction because the encoder has more capacity to preserve information from the input. In the basic fully connected study, MSE decreased from 0.040465 at dimension 2 to 0.008209 at dimension 32, while SSIM increased from 0.545584 to 0.905917. The larger latent space therefore reduces compression and preserves more image information.

Difference between fully connected and convolutional reconstruction. The convolutional autoencoder achieved MSE 0.001133 and SSIM 0.988807 compared with MSE 0.014010 and SSIM 0.843624 for the fully connected autoencoder. The convolutional model can exploit local spatial structure and shared filters, which is well matched to image data.

Effect of Gaussian noise level. As σ increased from 0.1 to 0.3, MSE increased from 0.002061 to 0.005750 and MAE increased from 0.013650 to 0.026295, while SSIM decreased from 0.975110 to 0.878240. Stronger corruption leaves the denoising model with less reliable information about the clean image.

Ability of the denoising autoencoder to recover image structure. For $\sigma = 0.2$, the denoising CAE achieved SSIM 0.962498. The qualitative plot shows that the principal digit strokes are recovered even after Gaussian corruption, demonstrating that the network has learned useful image structure rather than simply copying the noisy input.

Characteristics of the VAE latent distribution. The VAE represents each input using μ and $\log \sigma^2$ and samples a latent vector using the reparameterization trick. The latent points form a continuous distribution rather than a set of isolated deterministic codes. This regularization enables sampling and interpolation in latent space.

Structure of the two-dimensional VAE latent space. The latent-space plot shows regions associated with different digit classes, together with some overlap. Similar-looking digits can occupy neighboring regions because the representation captures visual features rather than explicit class labels.

Quality and diversity of generated samples. Randomly sampled latent vectors produce recognizable handwritten-like digits with visible variation. Some samples are ambiguous or less sharp, which is consistent with sampling from regions that may not correspond to a strongly represented training pattern.

Smoothness of latent-space interpolation. The interpolation figure shows gradual changes between the endpoint reconstructions. This indicates that nearby latent vectors are mapped to related image patterns and that the decoder has learned a relatively smooth latent-to-image mapping.

Relationship between reconstruction error and visual quality. Higher reconstruction error generally corresponds to more visible differences between the original and reconstructed images. The high-error examples illustrate that unusual or ambiguous handwritten structures can produce larger visual errors.

Deterministic versus probabilistic latent representations. A conventional autoencoder maps each input to one deterministic latent vector. The VAE instead learns parameters of a distribution and samples from it. The probabilistic formulation introduces a KL regularization term but provides a structured latent space that can be sampled and interpolated for generation.

26. Additional Latent-Dimension Study

The basic fully connected autoencoder was retrained using latent dimensions $\{2, 8, 16, 32\}$.

Latent Dimension	MSE	SSIM	Parameters
2	0.040465	0.545584	218514
8	0.019565	0.783788	219288
16	0.014794	0.832589	220320
32	0.008209	0.905917	222384

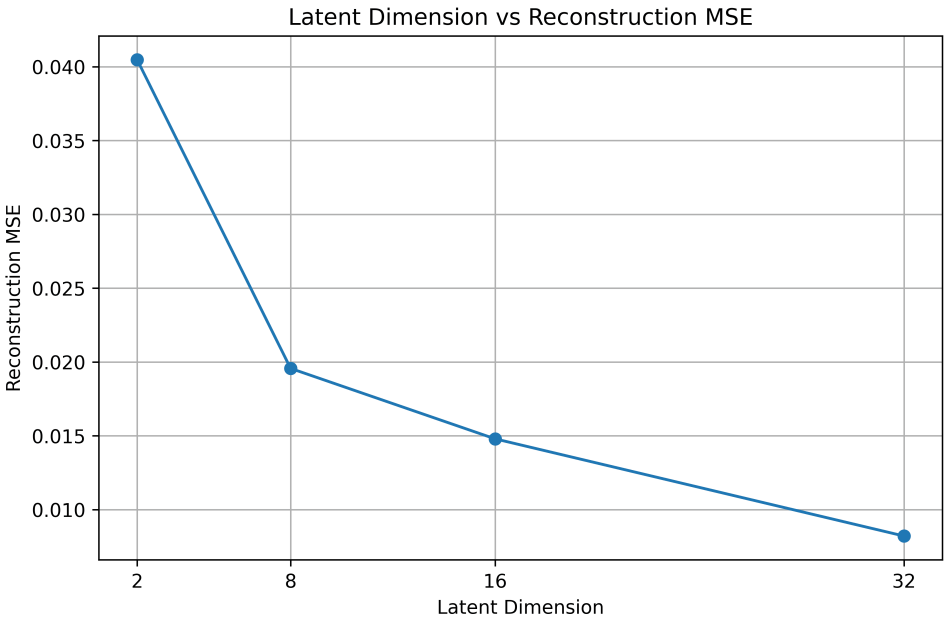


Figure 12: Latent dimension versus reconstruction MSE for the fully connected autoencoder.

Increasing the latent dimension reduces the compression bottleneck. Consequently, the model can preserve more information and the reconstruction MSE decreases. The improvement is substantial from dimensions 2 to 32, while the parameter count changes only slightly because the latent layer is a relatively small part of the complete network. The result illustrates the trade-off between compact representation and reconstruction fidelity.

27. Discussion Questions

1. **What is an autoencoder?** An autoencoder is a neural network trained to reproduce its input. It consists of an encoder that maps the input to a latent representation and a decoder that reconstructs the original input from that representation.
2. **What is the purpose of the encoder?** The encoder extracts useful features from the input and compresses them into a latent representation.
3. **What is the purpose of the decoder?** The decoder maps the latent representation back to the original data space and attempts to reconstruct the input as accurately as possible.
4. **What is a latent representation?** It is a compact numerical representation containing information that the model considers important for reconstructing the input.
5. **Why is the latent representation usually smaller than the input?** A smaller latent space creates a bottleneck that forces the network to learn informative features instead of simply copying every input value.
6. **What happens when the latent dimension is too small?** Too much information is discarded, so reconstruction error increases and fine details are lost. In the experiment, dimension 2 gave MSE 0.040465 compared with 0.008209 for dimension 32.
7. **What happens when the latent dimension is increased?** The model can retain more information, so reconstruction quality generally improves. The observed MSE decreased and SSIM increased as the latent dimension increased in the basic study.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?** Convolutional layers operate on local neighborhoods and reuse the same filters across the image. This allows them to learn edges, strokes and local spatial patterns directly.

9. **Differentiate an autoencoder and a denoising autoencoder.** A standard autoencoder reconstructs its input, whereas a denoising autoencoder receives a corrupted image but is trained to reconstruct the corresponding clean image.
10. **Why is the clean image used as the target in denoising?** The goal is to learn the underlying clean structure rather than reproduce the noise. The clean image provides the reference needed to learn what information should be removed.
11. **What happens when the noise level is increased?** Reconstruction quality decreases. In the Gaussian-noise experiment, MSE and MAE increased and SSIM decreased as σ increased from 0.1 to 0.3.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?** MSE penalizes squared pixel differences, MAE measures average absolute deviation, and SSIM evaluates structural similarity. Together they provide complementary views of reconstruction quality.
13. **What is a Variational Autoencoder?** A VAE is a generative autoencoder that learns a probability distribution over latent variables rather than assigning each input only one deterministic code.
14. **How does a VAE differ from a conventional autoencoder?** A conventional autoencoder learns a deterministic mapping $x \to z$, whereas a VAE learns distribution parameters such as μ and $\log \sigma^2$ and samples z from that distribution.
15. **Why does a VAE learn a distribution instead of a single latent vector?** The distribution makes the latent space regular and continuous enough to support meaningful sampling, interpolation and generation.
16. **What is the reparameterization trick?** It expresses sampling as $z = \mu + \sigma\epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. This separates the random component from the learnable parameters so gradients can pass through μ and σ.
17. **Why is ϵ sampled from a standard normal distribution?** It provides a fixed, parameter-independent source of randomness while allowing the learned μ and σ to control the resulting latent distribution.
18. **What is the role of KL divergence in the VAE loss?** KL divergence penalizes the learned latent distribution when it moves too far from the chosen prior. This regularizes the latent space and makes sampling from the prior more meaningful.
19. **What is the prior distribution used in the VAE?** The experiment uses the standard normal prior $p(z) = \mathcal{N}(0, I)$.
20. **Why is a two-dimensional latent space useful for visualization?** Each image can be represented directly as one point in a plane, making clusters, overlap and continuous transitions easy to inspect.
21. **What does latent-space interpolation demonstrate?** It tests whether moving smoothly between two latent vectors produces smoothly changing reconstructions. The experiment shows gradual visual changes between the selected endpoints.
22. **Why can a VAE generate new images?** Once the latent distribution has been regularized toward the prior, new latent vectors can be sampled and passed through the decoder to obtain new images.
23. **Why might some VAE-generated images be unclear?** A sampled latent point may not correspond to a highly populated region of the training latent space. The decoder can therefore produce an image that lies between common patterns or is weakly defined.
24. **What is the difference between reconstruction and generation?** Reconstruction starts with an existing image and attempts to reproduce it. Generation starts from a sampled latent vector and produces a new image without requiring a specific input image.
25. **Why should test images remain unseen during training?** Keeping the test set unseen provides an unbiased estimate of how well the learned model generalizes to new data.

28. Additional Exercises

1. **Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.**
   The measured results were:

Latent Dimension	MSE	SSIM	Parameters
4	0.038665	0.578751	65797
8	0.052944	0.385760	90889
16	0.022385	0.750616	141073
32	0.005710	0.938906	241441

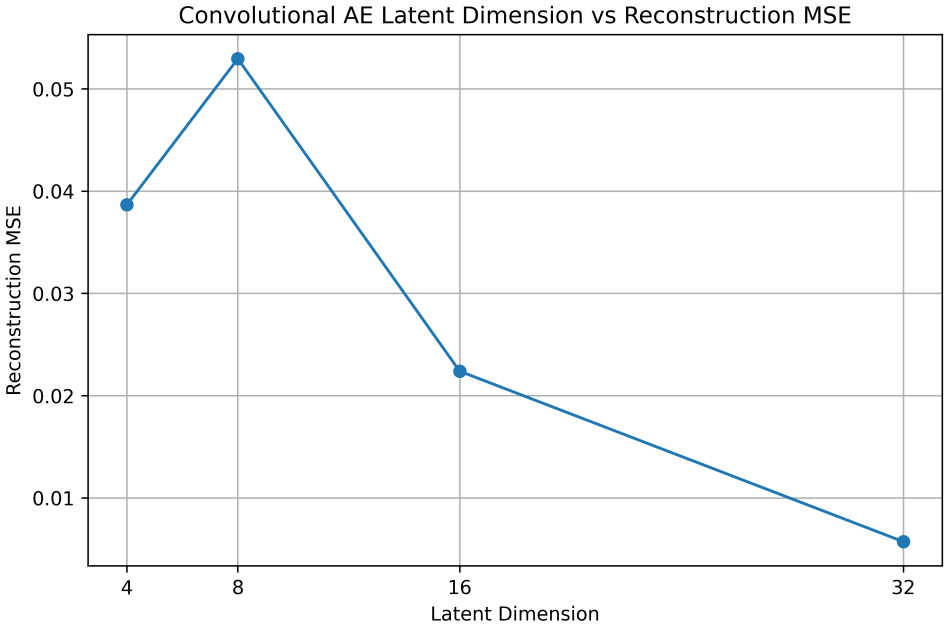


Figure 13: Convolutional autoencoder latent dimension versus reconstruction MSE.

The general trend is that the larger latent space gives better reconstruction, with dimension 32 giving the lowest MSE and highest SSIM. The dimension-8 result is not monotonic with dimension 4 in this particular run, which can occur because training is stochastic and the models were independently initialized. The main structural trend is recovered at dimension 16 and becomes especially clear at 32, where the bottleneck is much less restrictive.

2. **Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.**
   The same denoising architecture was evaluated under both corruption types. The Gaussian-noise results were:

Gaussian σ	MSE	MAE	SSIM
0.1	0.002061	0.013650	0.975110
0.2	0.003017	0.016957	0.962421
0.3	0.005750	0.026295	0.878240

The salt-and-pepper results were:

Probability	MSE	MAE	SSIM
0.05	0.004193	0.019843	0.912766
0.10	0.008260	0.031467	0.810776
0.20	0.020739	0.063739	0.644821

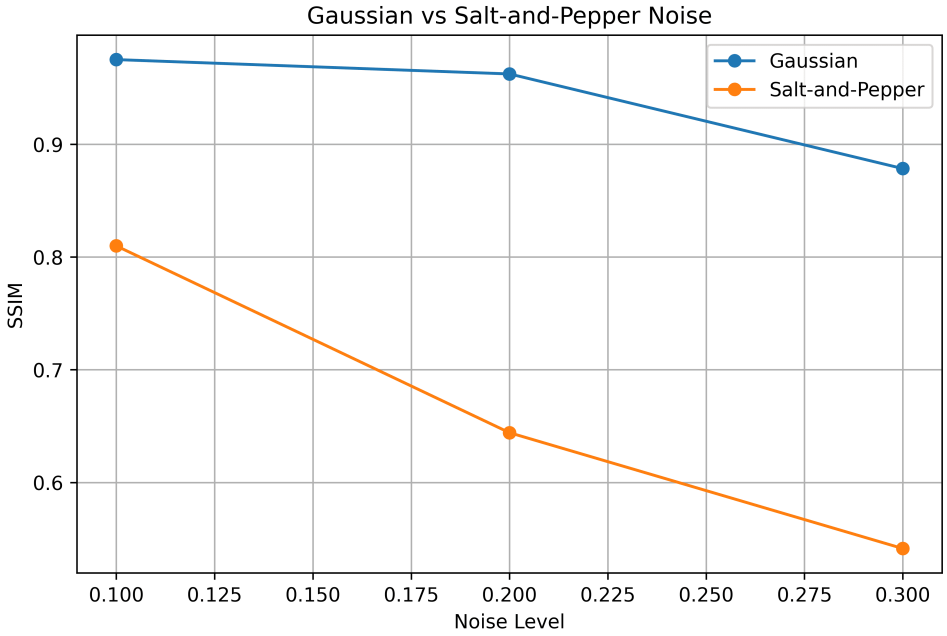


Figure 14: SSIM comparison between Gaussian and salt-and-pepper noise.

The salt-and-pepper corruption causes a stronger reduction in SSIM at comparable nominal noise levels. Its impulsive 0/1 replacements remove or overwrite pixels rather than making small continuous perturbations. Therefore, the denoising network has to infer missing pixel information more aggressively.

3. **Train the denoising autoencoder using two different noise levels and compare SSIM.**
   For Gaussian noise levels 0.1 and 0.3, the measured SSIM values were 0.975121 and 0.879561, respectively.

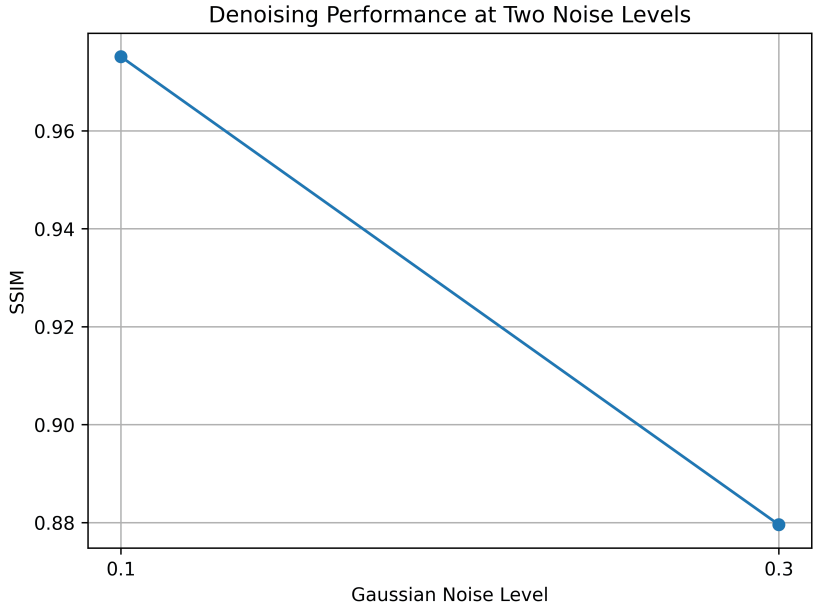


Figure 15: SSIM at two Gaussian noise levels.

The result confirms that stronger input corruption makes recovery more difficult. The network can remove moderate noise effectively, but as the noise variance increases, less reliable information remains in the input and structural similarity decreases.

4. **Replace the convolutional decoder with transposed convolutions and compare the reconstructions.**
   The transposed-convolution decoder produced MSE 0.025052, SSIM 0.730732 and 178001 parameters in this run.

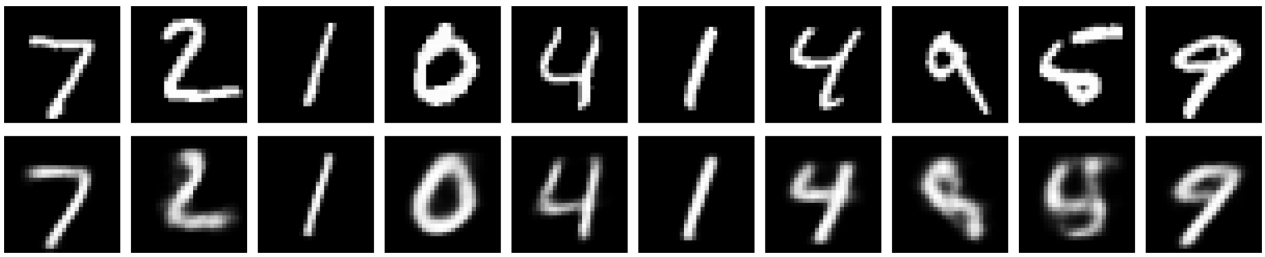


Figure 16: Reconstructions produced by the transposed-convolution decoder.

The result is weaker than the main convolutional autoencoder, which achieved MSE 0.001133 and SSIM 0.988807. This does not imply that transposed convolutions are inherently worse; rather, this particular architecture, parameterization and training run produced lower reconstruction quality. The comparison shows that decoder design and training configuration both affect the final reconstruction.

5. **Implement a VAE with latent dimension 2 and 8 and compare the latent representations.**
The two models produced latent tensors of shapes $(10000, 2)$ and $(10000, 8)$. The two-dimensional representation can be visualized directly as a plane, whereas the eight-dimensional representation cannot be displayed completely in 2D without a projection.

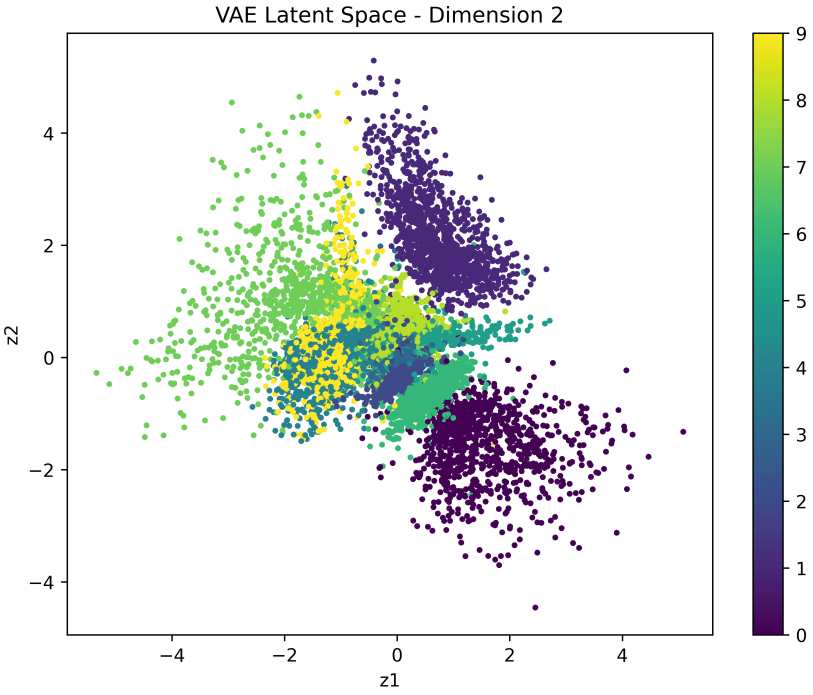


Figure 17: Two-dimensional VAE latent representation.

The dimension-2 model provides an immediately interpretable latent geometry, but it has limited capacity. The dimension-8 model can preserve more independent features because it has four times as many latent coordinates. The experiment's saved dimension-8 plot is a 2D visualization rather than a complete 8D visualization; therefore, the shape $(10000, 8)$ is the reliable quantitative result for the higher-dimensional representation.

6. **Investigate the effect of increasing the KL-loss contribution.**
The experiment varied β in the total objective $\mathcal{L} = \mathcal{L}_{rec} + \beta\mathcal{L}_{KL}$.

β	Reconstruction Loss	KL Loss	Total Loss
0.1	139.263229	10.416072	140.304840
1.0	141.365387	6.490149	147.855530
2.0	142.547272	5.524297	153.595871
5.0	148.363602	4.163291	169.180054

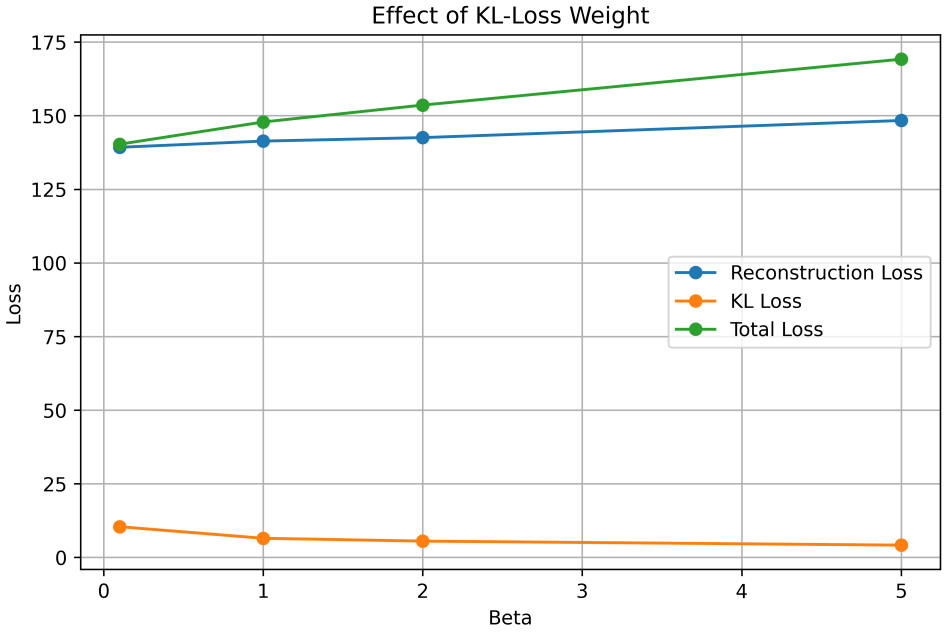


Figure 18: Effect of increasing the KL-loss weight β.

Increasing β places more emphasis on matching the latent distribution to the prior. In the observed run, KL loss decreases while reconstruction loss increases as β becomes larger. This is the expected trade-off: stronger latent regularization can improve the organization of the latent space, but it can also restrict the information available for detailed reconstruction.

7. **Generate a larger grid of VAE samples and discuss diversity.**

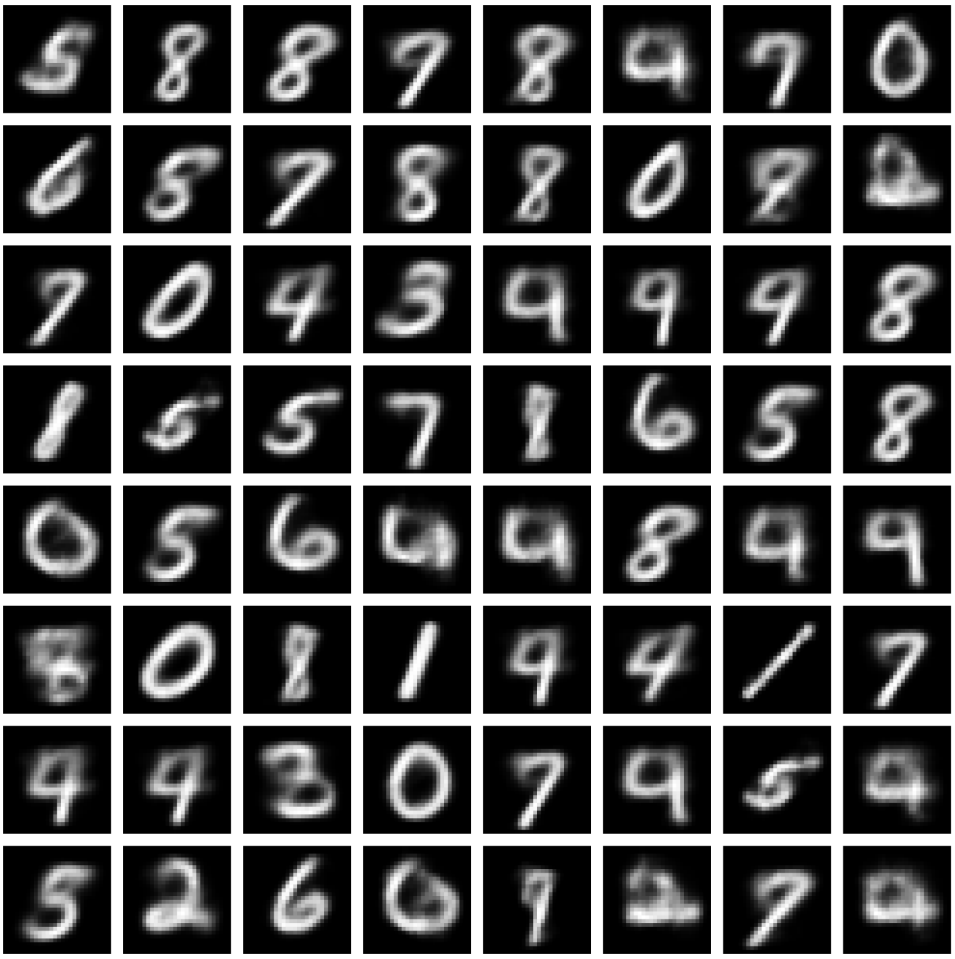


Figure 19: Sixty-four randomly generated VAE samples.

The larger grid makes the variation between generated samples easier to observe. Several samples resemble recognizable digits, while others are ambiguous or blurred. This variation is evidence that the decoder is not simply memorizing a single output pattern. At the same time, the presence of unclear samples shows that not every randomly sampled latent point corresponds to a highly realistic digit.

8. **Select two digit regions in the latent space and perform interpolation.**
   The experiment selected digits 3 and 8 and generated 11 interpolated latent vectors between their encoded representations.

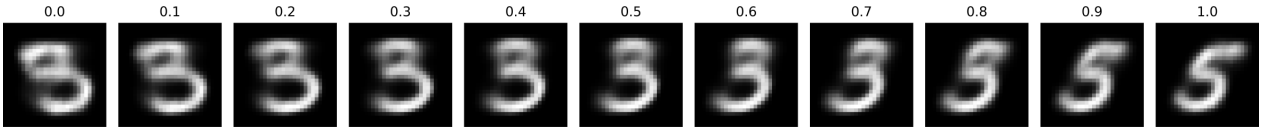


Figure 20: Interpolation between latent representations associated with digits 3 and 8.

The sequence shows a gradual transition between the two endpoint representations. This demonstrates that the latent space encodes more than isolated digit labels: nearby points can correspond to intermediate visual characteristics. Smooth interpolation is one of the main practical advantages of learning a structured continuous latent distribution.

29. Expected Outcome

The experiment demonstrates the progression from deterministic reconstruction to probabilistic generative modeling. The fully connected autoencoder learns a compact representation but loses some spatial detail. The convolutional architecture substantially improves reconstruction by exploiting image structure. The denoising autoencoder learns to recover clean digit structure from corrupted observations, with reconstruction quality decreasing as the noise level becomes stronger.

The VAE introduces a probabilistic latent representation. Its two-dimensional latent space shows class-related organization, while random sampling and interpolation demonstrate that the learned representation can be used for controlled generation. The quantitative results also show that generative regularization and reconstruction fidelity are related but distinct objectives.

30. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.
3. Pascal Vincent et al., "Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion," Journal of Machine Learning Research, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, "MNIST Handwritten Digit Database."
5. TensorFlow Documentation: `https://www.tensorflow.org`
6. Keras Documentation: `https://keras.io`
7. scikit-image Documentation: `https://scikit-image.org`

31. Source Code

The complete implementation of the Autoencoders and the other models developed for this experiment is available in the GitHub repository below. The repository contains the Jupyter Notebook, source code, dataset, and all generated results.

Repository Link

`https://github.com/AI-MANIAC007/Hands-On-Deep-Learning`